RWI-HC-010 (revision 2) — CFB Wainwright and the whole training area, 1 October to 2 November 2026

Ad hoc hindcast requested 2 Nov 2026. Box 52.50–52.95 N, 111.35–110.55 W: the garrison and the full training area. Sources: NASA FIRMS (all satellites, area API), Sentinel-2 / Sentinel-1 / Landsat passes over the box, NASA Black Marble night lights at the garrison, ERA5 weather, and the adsb.lol archive within 80 km of the garrison for every day of the window. **Times UTC, MDT (UTC−6) beside.** Revision 2 adds the public-reporting check (section 5) that revision 1 left out and revises the leads against it. LLM-assisted analysis.

1. Bottom line

1.0 **The 17 Oct fire killed four soldiers and hospitalised 39** (C Company, 3 PPCLI, live-fire training before a Latvia deployment). It started on the grenade-launcher range inside a permanent danger area that crews could not enter, crossed the fire guards and burned beyond the base; investigations by the military police, a Board of Inquiry and the Alberta Chief Medical Examiner are under way (section 5). Revision 1 of this hindcast did not say so; it should have.

1.1 **Three heat events in 33 days, all in the training area, nothing on the garrison.** 14 Oct (one pixel, 11.7 MW), 17 Oct (the range fire, 8 pixels up to 11.5 MW), and 18 Oct (two pixels, 3.9 MW, about 4 km south-east of the 17 Oct fire). No detection 1–13 Oct or 19 Oct – 2 Nov.

1.2 **A burn scar was already there before October.** The clear Sentinel-2 pass of 4 Oct shows a dark-red (SWIR) scar south-west of the central lake and a smaller one to its west. It predates the window; dating it needs September imagery (lead L2).

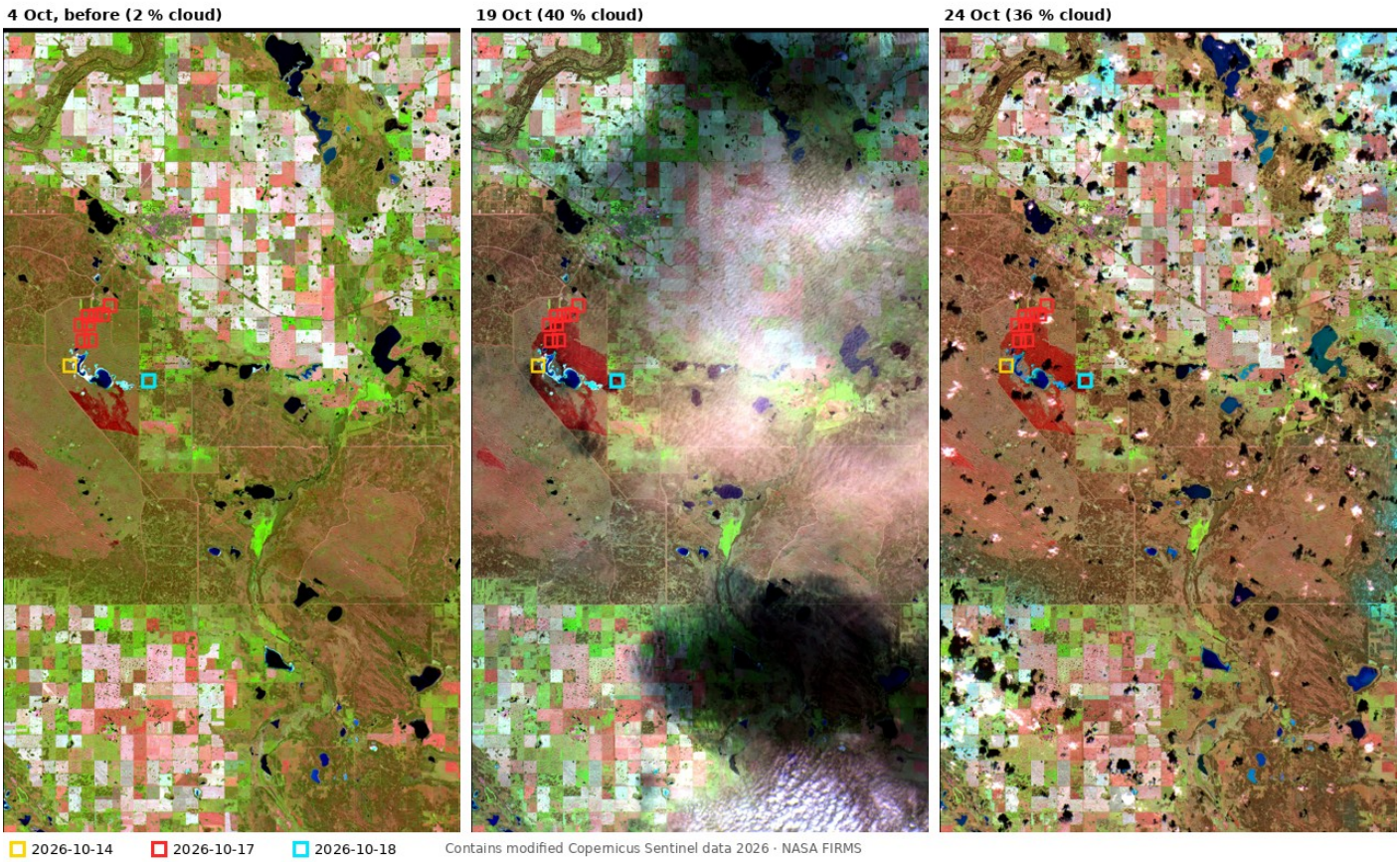
1.3 **The 17 Oct fire's scar is visible on 19 and 24 Oct** wrapping the lake (measured earlier at about 10.7 km²; BRIEF-002).

1.4 **No military aircraft landed or flew low near the garrison in the archive.** Every military aircraft within 80 km was at cruise altitude (24,000–37,000 ft) on the airways that cross the area: USAF C-17s, KC-135s, RCAF CC-144 Challengers, a CC-130J, a CC-330, a German Air Force Global Express (31 Oct) and the C-32A 98-0002 (callsign AF2, 7 Oct). Low-level coverage here is thin, so this does not rule out low flying; the CC-130H of the 17 Oct SAR jump had its transponder off (BRIEF-002).

1.5 **Night lights** at the garrison show no outage or surge beyond cloud and moonlight effects.

2. Heat (VIIRS, FIRMS area API)

Date	UTC (MDT)	Pixels	Max MW	Where	Reading
14 Oct	20:51 (14:51)	1	11.7	52.75 N 110.94 W, west of the 17 Oct fire	Separate fire, unreported (carried from BRIEF-002)
17 Oct	20:49 (14:49)	8	11.5	52.77–52.78 N 110.92–110.93 W	The range fire (SITREP 005, BRIEF-002)
18 Oct	20:12, 20:32 (14:12, 14:32)	2	3.9	52.75 N 110.87 W, ~4 km SE of 17 Oct	**New:** rekindle or a second fire the day after (L1)



P1 · The training area in shortwave infrared before (4 Oct) and after (19, 24 Oct) the 17 Oct fire, with every VIIRS detection of the window. The scar south-west of the lake on 4 Oct predates the window.

3. Imagery available in the window

Sensor	Passes (UTC date, scene cloud)	Use
Sentinel-2	4 Oct (2 %), 9, 14 (74 %), 16, 19 (40 %), 24 (36 %), 29 Oct	4 Oct clean baseline; 19 and 24 Oct partly clear after
Landsat 8/9	1 Oct (0 %), 8, 9, 16, 17 (83 %), 24 (26 %), 25 (28 %), 1 Nov	1 Oct clean baseline; 24–25 Oct after (30 m)
Sentinel-1 radar	2, 7, 14, 19, 26, 31 Oct (all ~01:05–01:13 UTC)	Cloud-free series; 14 vs 19 Oct brackets the fire

4. Flights within 80 km (adsb.lol archive), 1 Oct – 1 Nov

Category	Aircraft (closest approach to the garrison, altitude, UTC)
USAF airlift / tankers	C-17s 1, 5, 10, 12 (5 km, 34,000 ft), 13, 15, 22, 26 Oct (5 km, 34,000 ft); KC-135s 1, 10, 14, 17, 23 Oct; USN P-8A 5 Oct
RCAF	CC-177 1 and 5 Oct; CC-144 Challengers 144617–144620 19–23 Oct (3 km, 32,000–36,000 ft); CC-130J 130612 22 Oct (3 km, 24,000 ft); CC-330 15002 29–30 Oct (17 km, 33,000 ft); King Air trainers (MSTG) early Oct
Foreign military	RAF A400M ZM421 9 Oct (38 km); German Air Force Global Express 14+07 (GAF621) 31 Oct 16:50 UTC (10:50 MDT), 3 km, 32,750 ft
State / VIP	USAF C-32A 98-0002, callsign AF2, 7 Oct 05:49 UTC (23:49 MDT on 6 Oct), 27 km, 33,000 ft
Low flying or landings	None in the archive (coverage below ~5,000 ft is thin here)

5. Leads and open questions

L1 · 18 Oct second hot spot. Two pixels at 52.75 N 110.87 W, 14:12 and 14:32 MDT on 18 Oct, about 4 km south-east of the 17 Oct fire on the edge of the farm fields. Reporting says the fire crossed the guards and burned beyond the base, was 'being held' by 23:00 on 17 Oct and was later managed south of Wainwright; Code Orange was lifted about 13:35 on 18 Oct. These pixels, 40–60 minutes after that, may be the escaped fire still burning off-base or a rekindle; no reporting mentions activity on 18 Oct. No new scar is visible there on 24 Oct. Ask: the MD of Wainwright fire department and DND media relations whether the fire was active on the afternoon of 18 Oct.

L2 · The pre-October scar. Already burned by 4 Oct, south-west of the lake. No reporting found of a September fire, a 2026 fire ban or a prescribed burn (the last announced prescribed burn was April 2024). Next step: a September 2026 pull (Sentinel-2, FIRMS) to date it, then ask DND whether ranges burned before 17 Oct.

L3 · 14 Oct fire (carried from BRIEF-002): one 11.7 MW pixel at the west end of the lake, three days before the deadly fire, in the same dry spell (3.5 mm of rain in October). No reporting mentions it. It bears on what the base knew about fire risk on its ranges that week.

5b. Public reporting (checked 2 Nov 2026)

17 Oct fire. Began on the grenade-launcher range during routine training; in a permanent danger area (possible unexploded munitions), so it was monitored, not fought; crossed the fire guards and burned beyond the base; peak winds about 82 km/h. A grass fire about 1 km away was known for several hours before it ran 1 km in 30–60 seconds onto C Company, 3 PPCLI (84 soldiers) around 16:00 MDT. Code Orange about 18:00; four dead, 39 hospitalised (19 critical); Code Orange lifted about 13:35 on 18 Oct. Satellites saw fire at about 12:17 (Landsat) and 14:49 MDT (VIIRS), at least 3.5 h before the official 'about 4 p.m.' (Global News). Investigations: military police, a Board of Inquiry (9–12 months), the Alberta Chief Medical Examiner. About 10 km² burned (satellite estimate cited by CTV); no official figure. The MD of Wainwright said the fire was 'being held' by 23:00 on 17 Oct; no evacuation order.

Not in any reporting found: the 14 Oct and 18 Oct detections, any September fire, a 2026 fire ban or prescribed burn, allied units at Wainwright, or the German Air Force flight of 31 Oct (an overflight at 32,750 ft, not a visit).

Visits: Defence Minister McGuinty to the base and the hospital on 19 Oct (not the fire site); a vigil on 21 Oct. The RCAF Challengers at cruise altitude over Wainwright on 19–23 Oct match the period of ministerial and official travel, but the archive shows none landing there.

Parachutes were seen near the Wainwright Health Centre on the evening of 17 Oct during the medevac surge (CP24); the aircraft is not named in reporting (BRIEF-002 identified a CC-130H with its transponder off).

Sources: CP24 19, 22 and 24 Oct; Al Jazeera 23 Oct; Global News 20 Oct (satellite) and timeline; The Deep Dive 18 Oct; Forces News 19 Oct; CTV Winnipeg 24 Oct (CC-130H retirement); Global News 2024 (prescribed burn).

6. Method and limits

FIRMS area API over the full box for 1 Oct – 2 Nov 2026 (11 detections). History pull (scripts/history_pull.py) for imagery chips, night lights and ERA5 weather, with 17 Oct as the event date. Flights: scripts/adsb_history.py --profile point 52.80 N 110.90 W, 80 km, each day 1 Oct – 1 Nov. VIIRS sees fires only at overpass times (about 01:30 and 13:30 local); small or short fires between passes are missed. Community ADS-B reception at low level near Wainwright is sparse.